



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

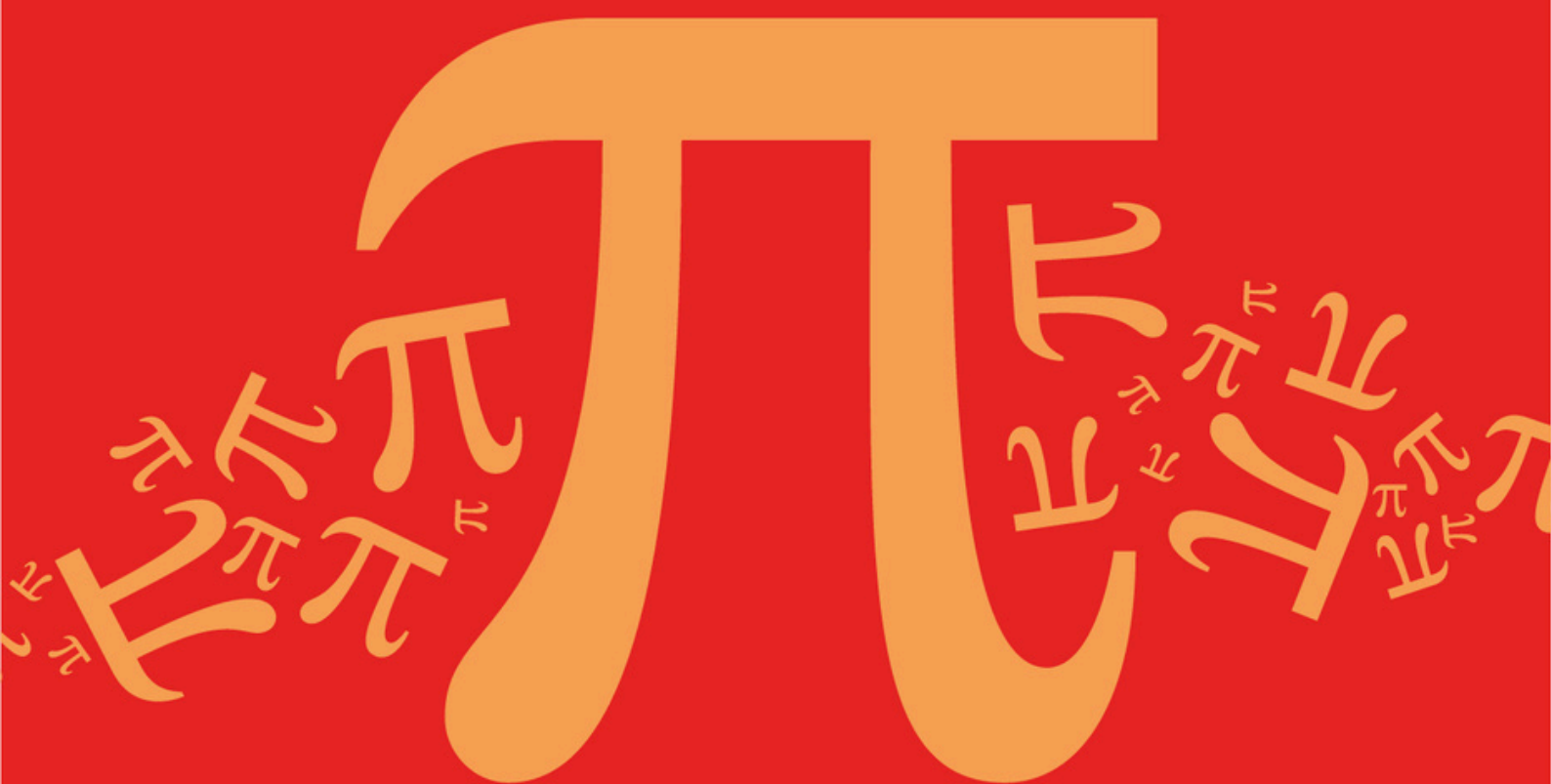
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C9: Factorise Determinants

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

C: Matrices

Sub topic

C9: Factorise Determinants

EXAM PAPERS PRACTICE

© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Explains that $\det \mathbf{M} = 0$ when $\mathbf{M}$ is singular (Seen anywhere)	AO2.4	R1	$\mathbf{S}$ is singular $\Rightarrow \begin{vmatrix} a & a & x \\ x-b & a-b & x+1 \\ x^2 & a^2 & ax \end{vmatrix} = 0$
Seeks factor by combining rows or columns to find a first linear factor for example $C_1' = C_1 - C_2$	AO3.1a	M1	$\det \mathbf{S} = \begin{vmatrix} 0 & a & x \\ x-a & a-b & x+1 \\ x^2-a^2 & a^2 & ax \end{vmatrix}$
Extracts first factor correctly	AO1.1b	A1	$= (x-a) \begin{vmatrix} 0 & a & x \\ 1 & a-b & x+1 \\ x+a & a^2 & ax \end{vmatrix}$
Combines rows or columns to find a second linear factor $R_3' = R_3 - aR_1$	AO1.1a	M1	$\det \mathbf{S} = (x-a) \begin{vmatrix} 0 & a & x \\ 1 & a-b & x+1 \\ x+a & 0 & 0 \end{vmatrix}$
Extracts second factor correctly	AO1.1b	A1	$= (x-a)(x+a) \begin{vmatrix} a & x \\ a-b & x+1 \end{vmatrix}$
Completes expansion and obtains final factor	AO1.1b	A1	$= (x-a)(x+a)(a+bx)$
Deduces correct values of x FT 'their' factors	AO2.2a	A1F	$(x-a)(x+a)(a+bx) = 0$ $x = a, -a, -\frac{a}{b}$
			Total 7 marks

Q2.

(a) eg $\begin{vmatrix} yz & x(y+z) & xy \\ x & y+z & z \\ x^2 & z^2-y^2 & z^2 \end{vmatrix}$

$$C_2' = C_2 + C_3$$

M1

$$= (y+z) \begin{vmatrix} yz & x & xy \\ x & 1 & z \\ x^2 & z-y & z^2 \end{vmatrix}$$

A1

2

$$(b) \quad \text{eg } (y+z) \begin{vmatrix} y(z-x) & x & xy \\ x-z & 1 & z \\ x^2-z^2 & z-y & z^2 \end{vmatrix}$$

$$C_1' = C_2 - C_3$$

Attempt at a second linear factor

M1

$$= (y-z)(y+z) \begin{vmatrix} -y & x & xy \\ 1 & 1 & z \\ x+z & z-y & z^2 \end{vmatrix}$$

A1

$$= (y-z)(y+z) \begin{vmatrix} -(x+y) & x & xy \\ 0 & 1 & z \\ x+y & z-y & z^2 \end{vmatrix}$$

$$C_1' = C_2 + C_2$$

Complete attempt at remaining factors (M0 if they just expand and can do nothing with it)

M1

$$= (x-z)(y+z)(x+y)(xz-xy-yz)$$

A1

©2025 Exam Papers Practice. All rights reserved.

4

[6]

Q3.

$$(a) \quad \Delta = \begin{vmatrix} 1 & 2 & 3 \\ x & y & z \\ x+y+z & y+z+x & z+x+y \end{vmatrix}$$

e.g.

$$R_3' = R_3 + R_2$$

M1

$$= (x+y+z) \begin{vmatrix} 1 & 2 & 3 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix}$$

A1

2

(b) Expanding remaining det.

M1

$$\Delta = (x + y + z)(x - 2y + z)$$

A1

2

[4]

Q4.

- (a) (i) Appropriate row/column operation
 eg $R_1' = R_1 + R_3$, $R_3' = R_3 + R_1$
 or $C_3' = C_3 - nC_2$

M1

$$\Delta = \begin{vmatrix} n^2 + n + 1 & 0 & 0 \\ 0 & 1 & n \\ 1 & -(n+1) & 1 \end{vmatrix}$$

$$\dots = (n^2 + n + 1) \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & n \\ 1 & -(n+1) & 1 \end{vmatrix}$$

Factor correctly extracted

A1

2

- (ii) Expanding remaining determinant

©2025 Exam Papers Practice. All rights reserved.

M1

$$\Delta = (n^2 + n + 1)^2$$

A1

2

- (b) $\Delta = (n^2 + n)^2 + 2n^2 + 2n + 1$
Accept unsimplified

B1

$$\dots = (n^2 + n)^2 + (n + 1)^2 + n^2$$

B1

2

- (c) Setting $n = 10$

M1

$$111^2 = 12321 = 110^2 + 11^2 + 10^2$$

A1

2

[8]

Q5.

(a) (i) $R_1' = R_1 + R_2$ or $R_2' = R_2 + R_1$

M1

leading to $R_{1/2} = (4 - q \quad 4 - q \quad 0)$

A1

2

(ii)

$$\Delta = (4 - q) \begin{vmatrix} 1 & 1 & 0 \\ -12 & -1 - q & -7 \\ 6 & 6 & 10 - q \end{vmatrix}$$

$$= (4 - q) \begin{vmatrix} 0 & 1 & 0 \\ q - 11 & -1 - q & -7 \\ 0 & 6 & 10 - q \end{vmatrix}$$

By $C_1' = C_1 - C_2$ (eg)

M1

$$= (4 - q)(q - 11) \begin{vmatrix} 0 & 1 & 0 \\ 1 & -1 - q & -7 \\ 0 & 6 & 10 - q \end{vmatrix}$$

©2025 Exam Papers Practice. All rights reserved.

2nd linear factor correct

A1

$= (4 - q)(q - 11) \times \dots$

Full attempt at 3rd factor

M1

$= (4 - q)(q - 11)(q - 10)$

A1

4

(b) (i)

$$\begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ -28 \end{bmatrix}$$

M1A1

$$= 4 \begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix} \text{ so that } \lambda = 4$$

A1

3

(ii) $\lambda = 10, 11$ noted or used
ft

B1

$$\begin{aligned} 0 &= 6x + 5y + 7z \\ \lambda = 10 \Rightarrow 0 &= -12x - 11y - 7z \\ 0 &= 6x + 6y \end{aligned}$$

B1

Using $y = -x$ to get $z \sim x/y$

M1

Eigenvector $(-7, 7, 1)$
Any non-zero multiple will do

A1

$$\begin{aligned} 0 &= 5x + 5y + 7z \\ \lambda = 11 \Rightarrow 0 &= -12x - 12y - 7z \\ 0 &= 6x + 6y - z \end{aligned}$$

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.
Either $y = -x$ or $z = 0$

M1

Eigenvector $(1, -1, 0)$
Any non-zero multiple will do

A1

7

(c) $\frac{x}{2} = \frac{y}{5} = \frac{z}{-7}, \quad \frac{x}{-7} = \frac{y}{7} = z$
Any line eqns.

M1

or $x = -y, z = 0$
any one ft correct

A1

2

Q6.

$$\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & xy \end{vmatrix}$$

$$= \begin{vmatrix} x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \\ yz & z(x-y) & y(x-z) \end{vmatrix}$$

By $C_2' = C_2 - C_1$ (eg)

M1

$C_3' = C_3 - C_1$ (eg)

M1

$$= (y-x)(z-x) \begin{vmatrix} x & 1 & 1 \\ x^2 & y+x & z+x \\ yz & -z & -y \end{vmatrix}$$

First two factors extracted (what's left has to be correct also)

A1A1

$$= (y-x)(z-x) \begin{vmatrix} x & 1 & 0 \\ x^2 & y+x & z-y \\ yz & -z & z-y \end{vmatrix}$$

By $C_3' = C_3 - C_2$ (e.g.)

M1

$$= (y-x)(z-x)(z-y) \begin{vmatrix} x & 1 & 0 \\ x^2 & y+x & 1 \\ yz & -z & 1 \end{vmatrix}$$

3rd factor extracted

A1

$$= (x-y)(y-z)(z-x)(xy + yz + zx)$$

Further R/C ops or expansion of remaining det (almost a dM1)

M1

CAO up to equivalents due to re-positioning of the signs

A1

Alternatives using *Cyclic Symmetry* and the *Factor Theorem* are fine

[8]

Q7.

(a) $D = x^2 + y^2 + z^2 - xy - yz - zx$

M1A1

2

(b) E.g. by $C_1' = C_1 + (C_2 + C_3)$

M1

$$\Rightarrow \Delta = \begin{vmatrix} x+y+z & y & z \\ 0 & z-x & x-y \\ 2(x+y+z) & y+x & z+y \end{vmatrix}$$

$$= (x+y+z) \begin{vmatrix} 1 & y & z \\ 0 & z-x & x-y \\ 2 & y+x & z+y \end{vmatrix}$$

Shown or explained from previous line

A1

2

(c) Working on (R/C-ops) or expanding

M1

remaining determinant

2nd factor = $-(x^2 + y^2 + z^2 - xy - yz - zx)$

Good attempt

dM1

$k = -1$

A1

3

[7]

Q8.

Expanding fully: $\Delta = x^3 + y^3 + z^3 - 3xyz$

B1

Using row/column operations:

eg $R_1' = R_1 + (R_2 + R_3)$

M1

$$\Rightarrow \Delta = (x+y+z) \begin{vmatrix} 1 & 1 & 1 \\ z & x & y \\ y & z & x \end{vmatrix}$$

With conclusion that $(x + y + z)$ is a factor of the required expression when $k = 3$

A1

NB Any line of argument that leads correctly from $(x + y + z) f(x, y, z)$ to $x^3 + y^3 + z^3 - 3xyz$ scores full marks

[3]

Q9.

(a)

$$\Delta = \begin{vmatrix} a-b & b & c \\ b-a & c+a & a+b \\ c(b-a) & ca & ab \end{vmatrix}$$

$$C_1' = C_1 - C_2$$

M1

$$= (a-b) \begin{vmatrix} 1 & b & c \\ -1 & c+a & a+b \\ -c & ca & ab \end{vmatrix}$$

©2025 Exam Papers Practice. All rights reserved.

A1

2

Or Setting $b = a \Rightarrow C_1 = C_2 \Rightarrow \Delta = 0$

Factor theorem

(M1)

$\Rightarrow (a - b)$ a factor of Δ

(A1)

(2)

Or $\Delta = (a-b)(c^3 + a^2b + ab^2 - abc - b^2c - a^2c)$

(M1)

Must be completely correct

(A1)

(2)

(b)

$$= (a-b) \begin{vmatrix} 1 & b-c & c \\ -1 & c-b & a+b \\ -c & a(c-b) & ab \end{vmatrix}$$

$$C_2' = C_2 - C_3$$

M1

$$= (a-b)(b-c) \begin{vmatrix} 1 & 1 & c \\ -1 & -1 & a+b \\ -c & -a & ab \end{vmatrix}$$

2nd linear factor extracted

A1

e.g. $\Delta = (a-b)(b-c) \begin{vmatrix} 0 & 0 & a+b+c \\ -1 & -1 & a+b \\ -c & -a & ab \end{vmatrix}$

Genuine attempt at both remaining linear factors: e.g. $R_1' = R_1 + R_2$

M1

and then expanding final det.
3rd factor

A1

$$\Delta = -(a+b+c)(a-b)(b-c)(c-a)$$

All correct

©2025 Exam Papers Practice. All rights reserved.

A1

5

Or By cyclic symmetry,
(b - c) and (c - a) are also factors

(M1)

(A1)

(A1)

Final linear factor & checking sign of a coefficient.

(M1)

(A1)

(5)

Or Expanding the determinant fully
 $\Delta =$

(M1)

(A1)

Multiplying out

$$(a - b)(b - c)(c - a)(a + b + c)$$

No fudging, or jumping straight to the answer allowed

(M1)

=

(A1)

Fully correct working to show the two things are identically equal & checking for sign

(A1)

(5)

[7]

Q10.

$$\Delta = \begin{vmatrix} y - x & x & x + y - 1 \\ x - y & y & 1 \\ y - x & x + 1 & 2 \end{vmatrix}$$

attempt at first linear factor, eg $C_1' = C_1 - C_2$

M1

$$= (y - x) \begin{vmatrix} 1 & x & x + y - 1 \\ 1 & y & 1 \\ 1 & x + 1 & 2 \end{vmatrix}$$

for 1st linear factor (Ignore remaining det.)

A1

$$\Delta = (y - x) \begin{vmatrix} 0 & x + y & x + y \\ 1 & y & 1 \\ 1 & x + 1 & 2 \end{vmatrix}$$

Attempt at second linear factor, eg $R_1' = R_1 + R_2$

M1

$$= (y - x)(y + x) \begin{vmatrix} 0 & 1 & 1 \\ 1 & y & 1 \\ 1 & x + 1 & 2 \end{vmatrix}$$

For 2nd linear factor (Ignore remaining det.)

A1

Full expansion

M1

$$\Delta = (y - x)(y + x)(2 - x - y)$$

Completely correct solution

A1

6

Or

$$\text{Setting } y = x \Rightarrow C_1 = C_2 \Rightarrow \Delta = 0$$

Factor theorem

(M1)

$$\Rightarrow (y - x) \text{ a factor of } \Delta$$

(A1)

$$\text{Setting } y = -x \Rightarrow R_1 = R_2$$

(M1)

$$\begin{aligned} \text{So that } R_1' &= R_1 + R_2 \Rightarrow R_1' = 0 \\ \Rightarrow \Delta &= 0 \text{ and } (y + x) \text{ a factor of } \Delta \end{aligned}$$

(A1)

Genuine attempt at 3rd factor

(M1)

Completely correct solution

(A1)

(6)

©2025 Exam Papers Practice. All rights reserved.

Additional notes:

M0 for full expansion from the start with no successful factorisation progress

M1 A1 M0 M1 A0 for full expansion after one factor found and remaining quadratic factor left unfactorised (or incorrectly done)

4 + M0 for two correct linear factors but final det incorrectly expanded.

5 + A0 for minor sign error, but correct otherwise

[6]

Q11.

$$(a) \quad \Delta = \begin{vmatrix} 0 & 0 & 1 \\ a-c & b-c & c \\ b(c-a) & a(c-b) & ab \end{vmatrix}$$

$$\text{By } C_1' = C_1 - C_3 \text{ and } C_2' = C_2 - C_3$$

M1

$$(a-c)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & c \\ -b & -a & ab \end{vmatrix}$$

1st and 2nd factors

A1

A1

$$= (a - c)(b - c)(b - a)$$

By expanding

M1

$$= (a - b)(b - c)(c - a)$$

AG

A1

5

(b) (i) Noting $a = b = 3, c = 5$ in coefft. matrix

M1

$$\Rightarrow \Delta = 0 \text{ since factor } (a - b) = 0$$

Or starting from scratch

Hence no unique soln. to system
Clearly explained

A1

2

(ii) $3R_2 + R_3 = 24R_1$

M1

©2025 Exam Papers Practice. All rights reserved.

giving consistency iff $24p - 3q - r = 0$

Answer given

A1

2

(iii) $x + y + z = 1$
 $3x + 3y + 5z = 8$
 $5x + 5y + 3z = 0$

$$(2) - 3 \times (1) \text{ or } (3) - 5 \times (1) \Rightarrow z = 2\frac{1}{2}$$

M1A1

Then $x + y = -1\frac{1}{2}$

B1

Setting $x = \lambda$ (e.g.) $\Rightarrow y = -1\frac{1}{2} - \lambda$
Parametrisation

M1

All correct, any form
CAO

A1

5

Special Case ruling for those who don't attempt to parametrise
but who show this *is* the full solution:

4 + B0

First alternative (a)

Setting $a = b$ (etc.) $\Rightarrow \Delta = 0$ since $C_1 = C_2$

M1

$\Rightarrow (a - b)$ a factor, by the factor theorem

A1

By cyclic symmetry, $(b - c)$ and $(c - a)$ are also

M1

factors

A1

Checking sign/coefft, = (+) 1

A1

5

©2025 Exam Papers Practice. All rights reserved.

Second alternative (a)

Expanding the determinant

M1

$$\Delta = ab^2 + a^2c + bc^2 - b^2c - ac^2 - a^2b$$

A1

Multiplying out $(a - b)(b - c)(c - a)$

No fudging, or jumping straight to the answer allowed

M1

$$= abc + ab^2 + a^2c + bc^2 - b^2c - ac^2 - a^2b - abc$$

A1

Fully correct working to show the 2 things =

A1

Q12.

$$\Delta = \begin{vmatrix} 1 & a^2 & bc \\ 0 & b^2 - a^2 & c(a-b) \\ 0 & c^2 - a^2 & b(a-c) \end{vmatrix}$$

Row/column operations (≥ 1 set)
By $R_2' = R_2 - R_1$ and $R_3' = R_3 - R_1$

M1 A1

$$= (b-a)(c-a) \begin{vmatrix} 1 & a^2 & bc \\ 0 & a+b & -c \\ 0 & c+a & -b \end{vmatrix}$$

Taking out factors (≥ 2 attempted)

M1 A1

$$= (b-a)(c-a)(c-b) \begin{vmatrix} 1 & a^2 & bc \\ 0 & a+b & -c \\ 0 & 1 & 1 \end{vmatrix}$$

By $R_3' = R_3 - R_2$ and then expanding fully
(including attempt at all four factors)

M1

$= (a-b)(b-c)(c-a)(a+b+c)$
Or by use of factor theorem and cyclic symmetry

Penalise "global" sign errors at the end only

A1

Q13.

$$(a) \quad \Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$$

By $R_1' = R_1 + R_2$

M1A1

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$$

Factoring out correctly

A1

$$\text{or } \begin{vmatrix} a+b+c & b & c \\ 2(a+b+c) & c+a & a+b \\ 0 & c-a & a-b \end{vmatrix}$$

By $C_1' = C_1 + C_2 + C_3$

$$= (a+b+c) \begin{vmatrix} 1 & b & c \\ 2 & c+a & a+b \\ 0 & c-a & a-b \end{vmatrix}$$

Etc.

Method for obtaining remaining factor

$$\Delta = 2(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

2 can be anywhere

M1

A1

5

(b) Nothing $a = a, b = 3, c = 1$

B1

Setting $\Delta = 0$

$$\text{i.e. } 0 = 2(a+4)(a^2-4a+7) \Rightarrow a = -4$$

M1

A1

Showing $(a-2)^2 + 3 \neq 0$ so only one value of a

Ignore incorrect quadratic factors until here

CSO

M1

A1

5

Alt.

$$\begin{vmatrix} a & 3 & 1 \\ 4 & 1+a & a+3 \\ 2 & 1-a & a-3 \end{vmatrix} = 2a^3 - 18a + 56$$

(B1)

Equating to zero + solving attempt

(M1)

$$2(a + 4)(a^2 - 4a + 7) = 0 \Rightarrow a = -4$$

(A1)

$$(a - 2)^2 + 3 > 0 \Rightarrow \text{no other real solutions.}$$

Or discriminant < 0

(M1A1)

(5)

[10]



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Examiner reports

Q2.

The algebraic manipulation of determinants has always been found tricky by at least half of all students in each MFP4 paper set thus far, and this trend continued this summer also.

Part of the difficulty here is in getting students not to expand prematurely; some students just cannot seem to resist doing so immediately, and many others expanded immediately after completing part (a), usually correctly. Giving them one factor, and requiring that they show it is a factor, not only gives them a bit of a start but forces them to do the manipulation. It was disappointing to see how many students, having successfully managed the first factor, lacked the confidence to repeat the process with their own choice of a second. Leaving the expansion until the last possible moment is the only way, for many, to get the complete factorisation correct.

Q3.

This was a straightforward starter to the paper, and was generally found to be so by candidates. However, almost half of all candidates failed to score all 4 marks, due largely to arithmetical slips and carelessness with signs. This was mostly due to unhelpfully lengthy work on row- and/or column-operations after the $(x + y + z)$ factor had been extracted, when the simplest approach was to just go ahead and expand the remaining determinant. This highlights the view that many of these candidates had had insufficient practice with handling determinants algebraically in order to arrive at the point when they might have a “feel” for what is a good approach to take at different stages. Moreover, even in this example, it was about 50-50 whether candidates added R_3 into R_2 or vice versa; the latter approach clearly producing a simpler determinant to continue to work with afterwards, if some thought had been given to it.

Q4.

Most candidates were successful in part (a), though some used more efficient methods than others. Many made a good start in part (b) but were unable to express $2n^2 + 2n + 1$ as the sum of two squares, thus losing the last three marks in the question. Some candidates seemed to be unaware that the ‘squares’ asked for here had to be the squares of polynomials, or in part (c) the squares of integers.

Q5.

The structure of these questions proved very helpful to candidates, and those who were reasonably careful could generally score at least 16 of this question’s 18 marks without any difficulty.

Q6.

As usual, the general treatment of determinants using row/column operations was variably received: those who could, did ... and generally scored at least six of the eight marks available. Those who were less clear about the topic generally scored nothing or, at best, scrabbled around for a couple of method marks somewhere. Fortunately, the proportion of candidates in the former camp continues to grow year-by-year.

A major bug-bear for the examiners is the decreasing inclination among candidates to give any hint as to what they are attempting to do with the rows and/or columns, leaving it to the markers to figure it out for themselves. In simple cases such as this, it isn’t too

much of a problem, but in more difficult cases examiners may be unable to award any marks unless there is a **clear method**.

Q7.

Throughout all the MFP4 papers set thus far, this topic of determinants has proved to be a problem to many candidates. It is clear that while some candidates *have* seen how to perform row and column operations on determinants, others equally clearly have not.

The phrase “expand the determinant” in part (a) was supposed to direct candidates away from using row and/or column operations, but many did not take the hint. Conversely, the set-up in part (b) obviously requires row/column operations to begin with but, once a linear factor has been extracted (the one given), candidates should have been able to spot that they can then expand directly, as they only have a quadratic expression to deal with. Some candidates went on to use row/column operations very concisely and successfully, but they were very much the exception rather than the rule.

Q8.

Because of its slightly unusual nature, this short question was put last, so that candidates would not spend time they could better employ elsewhere in trying to figure out what to do with it. In the event, it proved far less of a problem than anticipated. The intention was that candidates would expand the determinant fully to gain the expression $x^3 + y^3 + z^3 - 3xyz$ and then use row/column operations to extract the factor of $(x + y + z)$. Most did indeed do so. The mark most commonly lost was the final one of tying the two ends together, which many failed to do, and examiners were quite strict in not giving full marks for solutions which never got round to joining up two otherwise disparate bits of working.

Q9.

This was the type of question previously handled very badly, but this time most candidates appeared well-drilled in how to manipulate determinants. The greatest obstacles to a completely successful conclusion to the question lay in the widespread instinct to expand the remaining determinant at too early a stage, leaving many candidates unable to factorise a quadratic expression in two variables, for instance; and in the more minor error of assuming that the final expression was fully cyclically symmetric in a, b, c , but failing to check that there was a ‘minus sign’ difference between what they were expecting and what they had been given.

Q10.

The manipulation of determinants continues to be a problem area for many candidates. Expanding from the outset is not really a good idea at all, and all who did so failed to cope satisfactorily with the resulting cubic expression. Even amongst those who found one linear factor before expanding, it was almost invariably the case that they were unable to cope with the resulting quadratic factor. This is very disappointing in its own right, especially when they often had a difference of two squares expression clearly written out in front of them.

Many candidates who used the expected row/column operations approach seemed ill-inclined to state anywhere what operations they were doing. This leaves the markers the task of guessing or deciphering the intended approach, and marks are not credited if the working proves too obscure to figure out.

Q11.

This question was a good source of marks for most candidates, although few managed to achieve full marks on it. Only around 50% of candidates knew how to manipulate determinants using row and/or column operations, although many got by on a much simpler level by expanding the determinant out to begin with, multiplying out the three brackets $(a - b)(b - c)(c - a)$, and showing that they gave the same expression. This is a valid and acceptable approach, even though not the intended one. Even so, many thought they could get away with expanding the determinant and then just saying that this was equal to the given answer $(a - b)(b - c)(c - a)$. They got two of the five marks available here. For future reference, it should be noted that handling determinants of higher order than a cubic is very unlikely to be done successfully by candidates using alternate approaches such as those described here and that straightforward row and column operations will be required to extract factors.

Answers to parts (b)(i) and (ii) were often poorly explained, despite being relatively straightforward in principle. A pleasing number of candidates realised that they could use the answer to part (a) in part (b)(i). Three of the five marks for part (b)(iii) were straightforward and most candidates got as far as $z = 2.5$ and $x + y = -1.5$.

Disappointingly, an overwhelming majority stopped here, failing to write the solutions in any obviously acceptable parameterised or vector line form, such as $(x, y, z) = (t, -1.5 - t, 2.5)$ or equivalent. Many, implicitly or explicitly, suggested that $x + y = -1.5$ is the equation of the line solution. Any candidates who explained why the answer $z = 2.5$ and $x + y = -1.5$ will suffice would, of course, have scored full marks, but satisfactory explanations were very few and far between.

Q12.

This was one of the questions for which many candidates seemed unprepared. Many showed no ability in manipulating determinants by using row or column operations. The number of candidates attempting to factorise the determinant by the factor theorem was very small. A few candidates gave the correct answer with little or no working, and these were given the benefit of the doubt. However, it should be noted that a direct approach to factorising determinants becomes less feasible as the degree of the terms increases. Only very capable students are likely to be able to use such methods successfully, although it is encouraging and impressive to see them do so.

Q13.

This question was less well done than Question 7 for the most part. Many candidates knew how to expand a determinant, but were unable to manipulate one using row/column operations. With part (a) proving to be difficult, part (b) was usually even more difficult. Candidates often failed to use part (a) to answer part (b). Very few candidates produced a second quadratic factor for the determinant and only a minority of these justified that the required value of a was unique.